Measurement of Inequality and Discrimination

Week 2: Causal Graphs

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Today

1. Causal graphs (Morgan & Winship, 2014, ch. 3)
2. Seeing and doing: probability distributions with and without intervention
3. The back-door criterion (Morgan & Winship, 2014, §§4.1–4.2)
4. Questions about the readings
5. Simpson's paradox: the Berkeley admissions data
6. Next week: Hu and Kohler-Hausmann (2025)

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§3.1 Identification

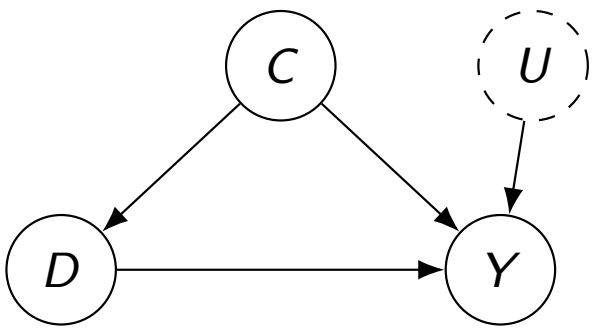
Two problems of causal inference (Morgan & Winship, 2014):

1. **Statistical inference**: how good are the estimated population parameters?
2. **Identification**: could we compute the causal effect given high-quality data about the population?



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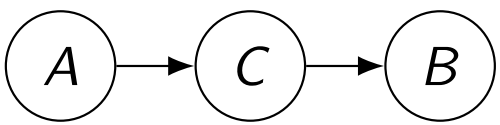
§3.2.1 The elements of a causal graph



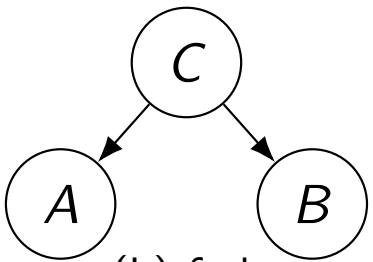
- **Nodes**: variables.
- **Arrows**: causal relations.
- **Paths**: any chain of arrows.
- **Directed paths**.
- **Parents/children**.
- **Ancestors/descendants**.

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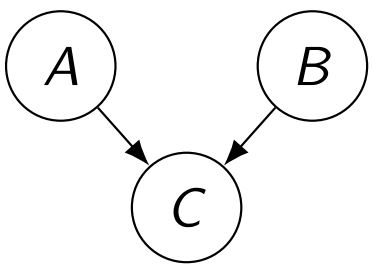
§3.2.2 Three basic patterns



(a) chain
(mediation)



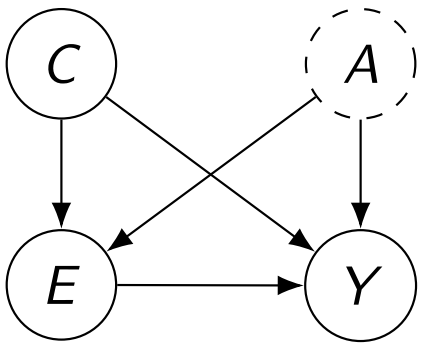
(b) fork
(common cause)



(c) collider
(common effect)

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§3.2.3 Confounding and back-door conditioning



E = education, Y = earnings, C = family background, A = ability. Morgan and Winship (2014, fig. 3.5)

- C and A are **confounders**.
- **Back-door conditioning** on C: compare people with the same family background.
- This removes the confounding by C, but not A (unobserved).

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A population as a table of proportions

Imagine a population of 1,000 people:

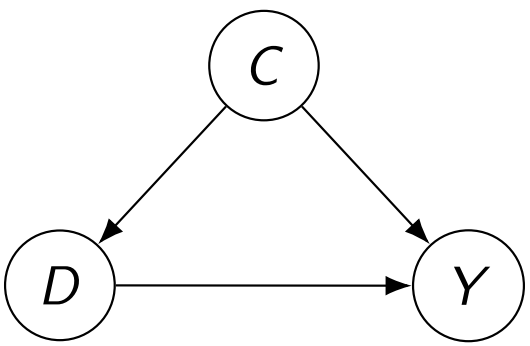
- C: wealthy family background ($C = 1$) or not ($C = 0$)
- D: has a university degree ($D = 1$) or not ($D = 0$)
- Y: has a high income ($Y = 1$) or not ($Y = 0$)

C	D	people	of whom $Y = 1$	share with $Y = 1$
1	1	400	280	70%
1	0	100	50	50%
0	1	100	40	40%
0	0	400	80	20%
		1,000	450	45%

The **probability distribution** of the population gives the share of people with each combination of values. For example,
$\Pr(Y = 1) = 450/1000 = 0.45$, and
$\Pr(C = 1, D = 1, Y = 1) = 280/1000 = 0.28$.

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Seeing: conditional probabilities



Conditioning means looking only at a subgroup.

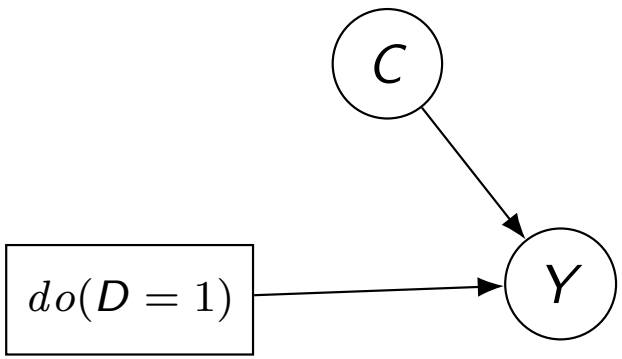
- $\Pr(Y = 1 \mid D = 1)$ is the share with a high income *among those with a degree*:

$$\frac{280 + 40}{400 + 100} = \frac{320}{500} = 64\%$$

- $\Pr(Y = 1 \mid D = 0) = \frac{50+80}{100+400} = \frac{130}{500} = 26\%$
- Difference: 38 percentage points.

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Doing: intervening on D



The "mutilated" graph: the arrow $C \to D$ is removed (Morgan & Winship, 2014, fig. 3.8).

- Imagine we *give everyone* a degree: $do(D = 1)$.
- This creates a different, hypothetical population.
- $\Pr(Y = 1 \mid do(D = 1))$ is the share with a high income in *that* population.

Seeing versus doing

- **Seeing**: $\Pr(Y \mid D = 1)$ describes the actual population.
- **Doing**: $\Pr(Y \mid do(D = 1))$ describes a hypothetical population. Causal effects are defined with it, e.g.:

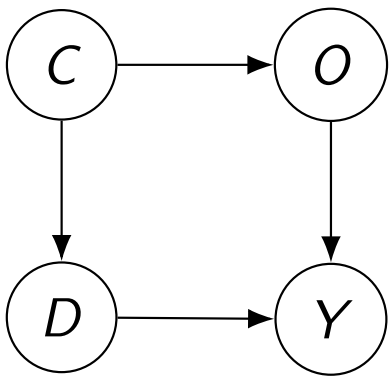
$$\mathrm{E}[Y \mid do(D = 1)] - \mathrm{E}[Y \mid do(D = 0)]$$



The two coincide when there are no back-door paths from D to Y.

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§4.1.1 Back-door paths



Morgan and Winship (2014, fig. 4.1)

- A **back-door path** from D to Y is a path that starts with an arrow *into* D: $D \leftarrow C \rightarrow O \rightarrow Y$.
- Back-door conditioning: "block" all back-door paths.
- Conditioning on C, on O, or on both blocks the back-door path.

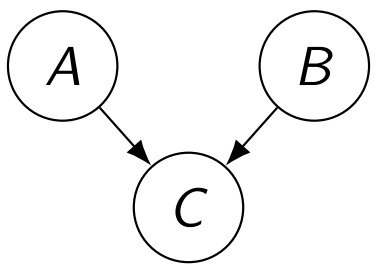
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§4.2 The back-door criterion

The back-door criterion: **block every back-door path, do not open new ones, and leave the causal path of interest alone.**

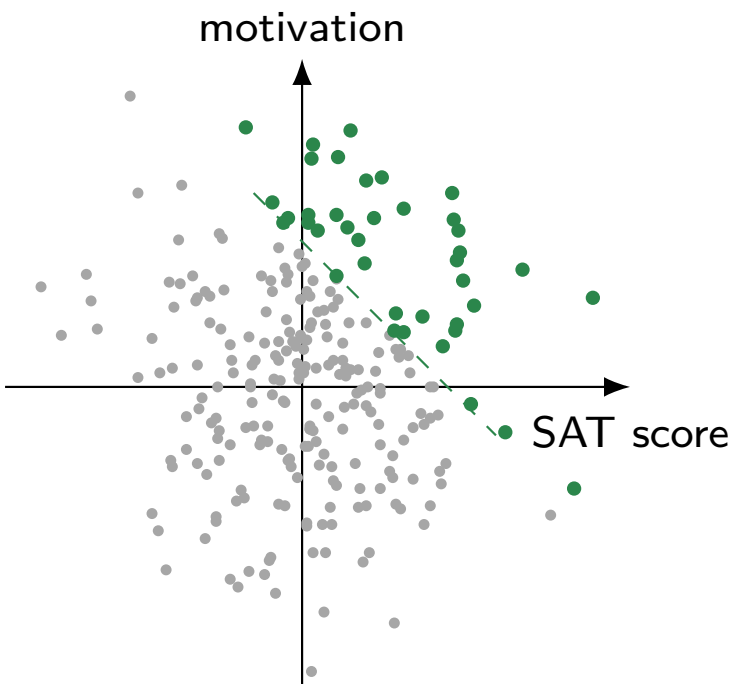
A path is **blocked** iff it contains

- a chain $\rightarrow C \rightarrow$ or a fork $\leftarrow C \rightarrow$ with C conditioned on, **or**
- a collider $\rightarrow C \leftarrow$ where neither C nor any descendant of C is conditioned on;



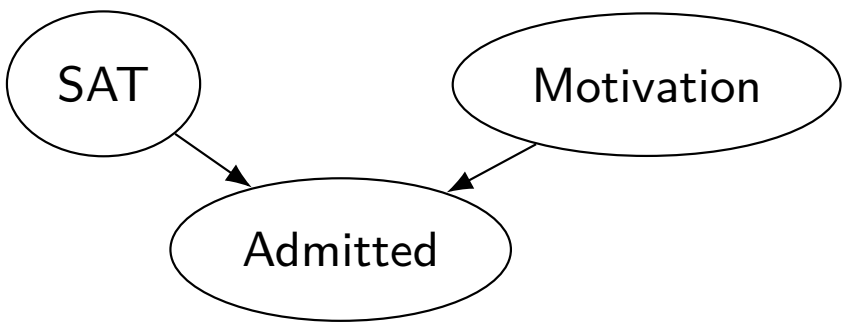
Collider

§4.1.2 Conditioning on a collider



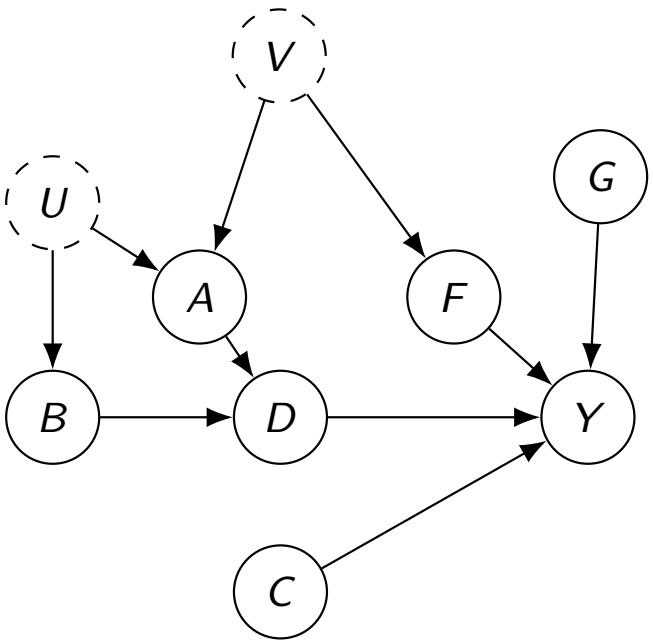
250 simulated applicants; green = admitted

A college admits applicants based on SAT score *and* motivation:



Questions about Morgan and Winship (2014), chapter 3 and sections 4.1–4.2

Exercise



after Morgan and Winship (2014, fig. 4.4)

We want the effect of D on Y.

1. List the back-door paths from D to Y.
2. On which path is A a collider?
3. Does conditioning on F identify the effect?
4. Does conditioning on A alone?
5. What about A, B, and F together?

Answers are in §4.2 of the reading.

Berkeley, 1973

Applications for graduate study at the University of California, Berkeley, fall 1973 (Bickel et al., 1975, table 1):

	applicants	admitted	share admitted
Men	8,442	3,738	44%
Women	4,321	1,494	35%

? Does Berkeley discriminate against women?

Department by department

	Men		Women	
Department	applicants	admitted	applicants	admitted
A	825	62%	108	**82%**
B	560	63%	25	**68%**
C	325	**37%**	593	34%
D	417	33%	375	**35%**
E	191	**28%**	393	24%
F	373	6%	341	**7%**
All six	2,691	**45%**	1,835	30%

Six largest departments; `UCBAdmissions` data from Bickel et al. (1975).

The adjustment formula

	All applicants		Share admitted	
Department	number	share	men	women
A	933	21%	62%	82%
B	585	13%	63%	68%
C	918	20%	37%	34%
D	792	17%	33%	35%
E	584	13%	28%	24%
F	714	16%	6%	7%

Adjustment formula: weight each department's admission rate by its share of *all* applicants, and add up:

$$\Pr(Y \mid do(G = \text{male})) =^{?} \sum_x \Pr(Y \mid G = \text{male}, X = x) \Pr(X = x)$$

Men: $0.21 \cdot 62\% + 0.13 \cdot 63\% + \ldots + 0.16 \cdot 6\% = 38.7\%$

Women: $0.21 \cdot 82\% + 0.13 \cdot 68\% + \ldots + 0.16 \cdot 7\% = 43.0\%$

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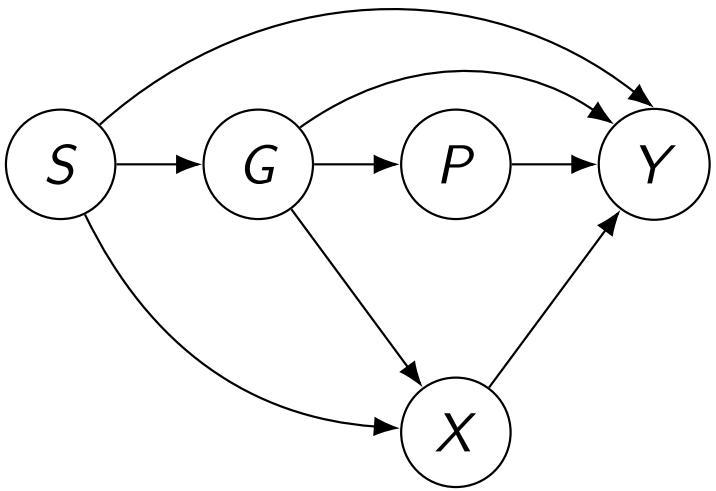
Simpson's paradox (Pearl, 2009, §6.1)

👉 **Adjusting for department**:

men 38.7%, women 43.0%

- **Simpson's paradox**: an association in the whole population is reversed in the subpopulations.
- Pearl: this is paradoxical only if we read the probabilities causally.

A causal graph for Berkeley

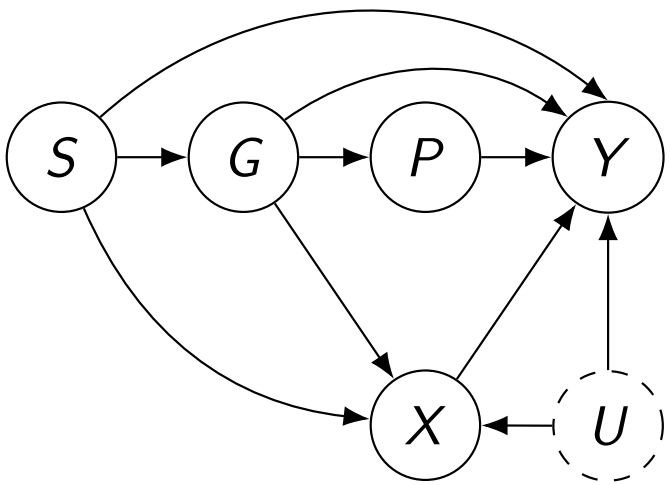


- S: biological sex
- G: gender assignment
- P: the applicant's gender as perceived by the college
- X: department applied to
- Y: admitted or not

? 1. Which paths are discrimination? Which paths may reflect unfairness?
2. Given the answer to 1, what are the back-door paths?

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What if the graph is wrong? (Pearl, 2009, §4.5.3)



U = aptitude

Pearl's worry: suppose aptitude affects both department choice and admission.

- Now X is a **collider** on $G \to X \leftarrow U \to Y$.
- Comparing within departments conditions on X and *opens* this path.

Hu and Kohler-Hausmann (2025)

Hu and Kohler-Hausmann (2025): "*What is perceived when race is perceived and why it matters for causal inference and discrimination studies*"

- Their **question**: what is a perception of race a perception *of*?
- **Claim 1**: "perception of race and perception of other decision-relevant features of an action situation are often co-constituted" (p. 239).
- **Claim 2**: which counterfactual comparison measures discrimination is a **normative** question.

An example

A prosecutor decides to file charges against a Black arrestee with **2 prior arrests**.

? Did the prosecutor's perception of the arrestee's race cause the decision?

What exactly do we change to make him "white"?

- Change only his skin colour and facial features?
- Change his skin colour and facial features *and* his number of prior arrests (say, to 1)?

Four questions about race

1. What perceptions are entailed in the treatment "perception of race"?
2. What cues trigger perception of race?
3. What makes someone a member of a particular racial category?
4. What is race?

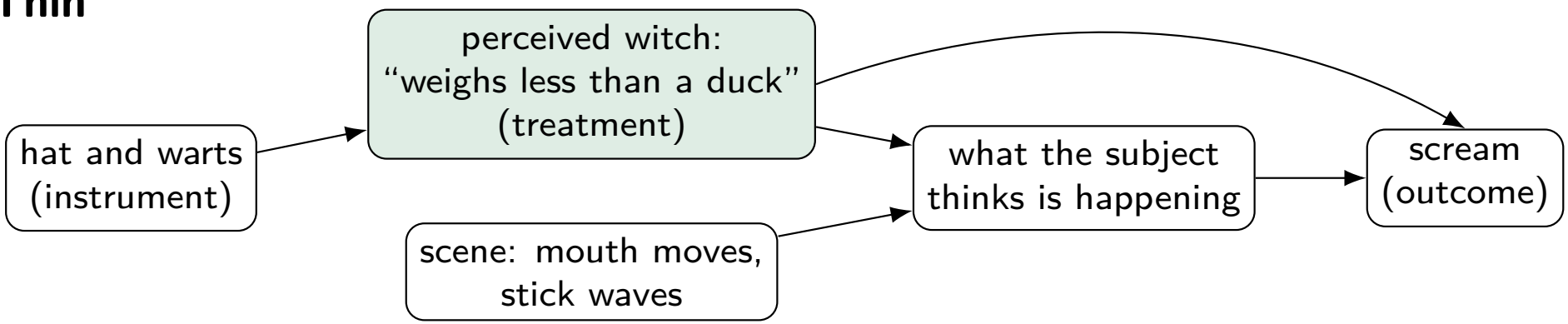
Hu and Kohler-Hausmann (2025, pp. 246–47): a causal study of race perception must answer (1) and (2), and these answers must be grounded in an answer to (4). Question (3) can be left open.

Witches (Hu & Kohler-Hausmann, 2025, pp. 247–48)

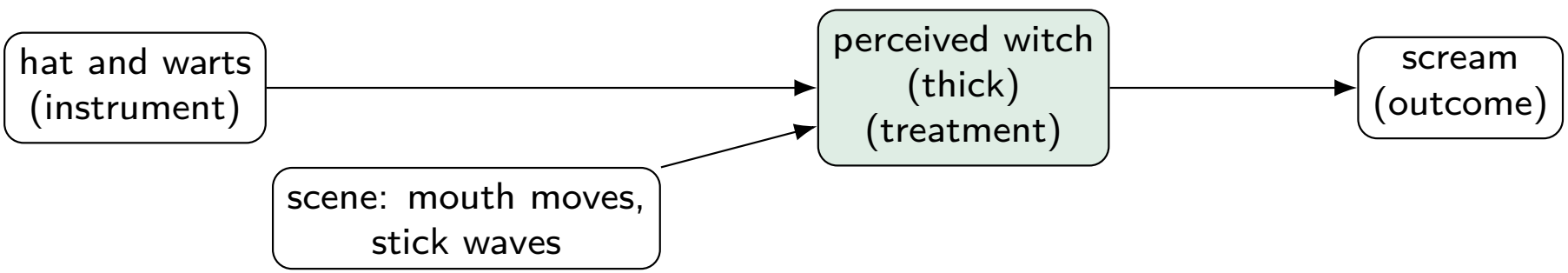
- A society divided into "witches" and "muggles". Witches are believed to have occult powers and to fly on brooms.
- Officially, a witch is someone who weighs less than a duck. Pointy hats and warts are signs of witchhood, but neither necessary nor sufficient.
- The anthropologist Ursula runs an experiment. Subjects watch a woman who moves her mouth and waves a small stick.
- The woman either wears a pointy hat and has warts, or wears a flat hat and has no warts.
- Outcome: does the subject scream?

Witches: thin and thick perception

Thin



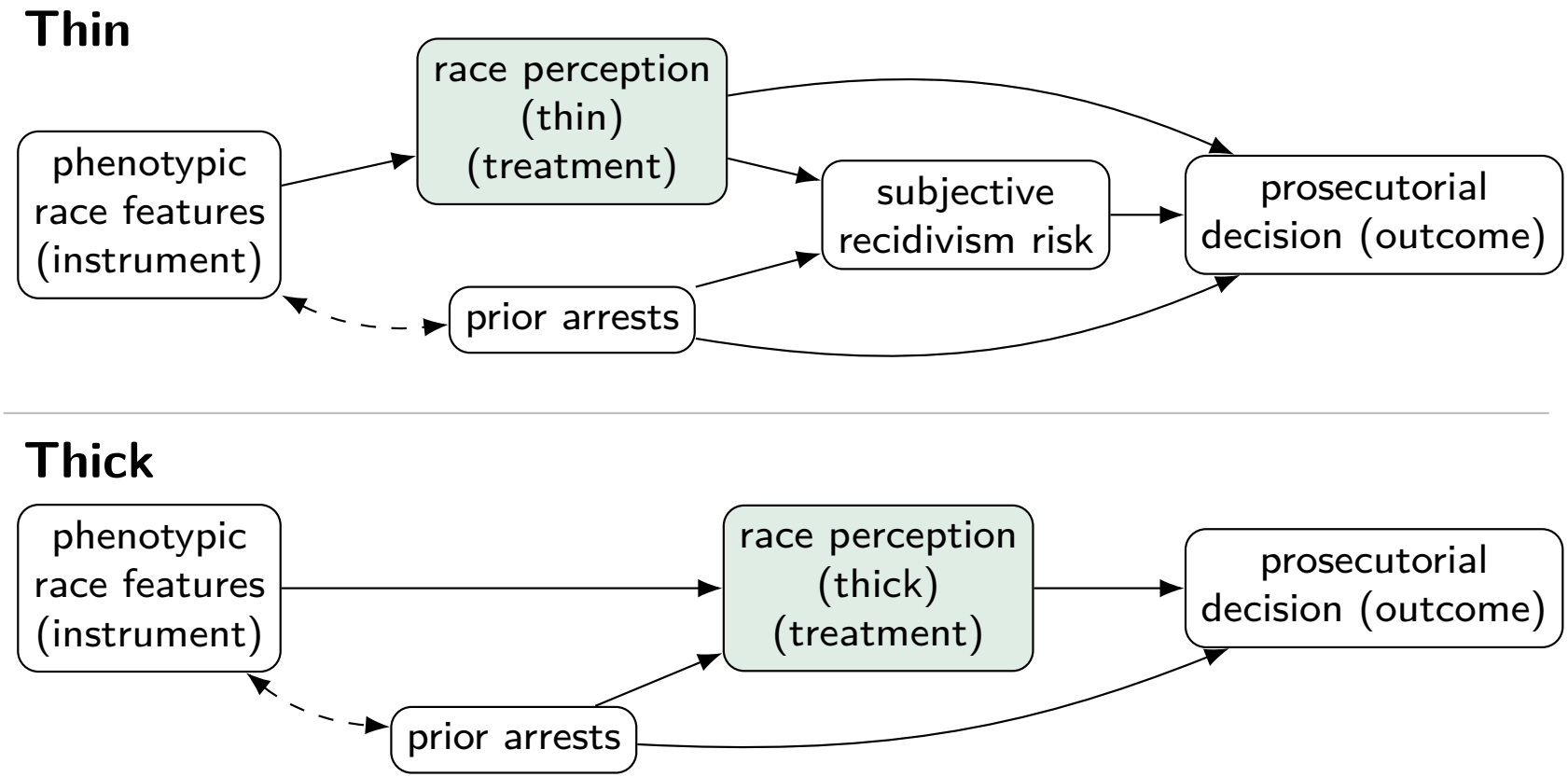
Thick



A thick view of race perception

- Critical race theory: race is socially constructed, grounded in social relations of inequality.
- On CRT, perceiving someone's race means perceiving them through a set of associations, meanings and stereotypes.
- Evidence from social psychology:
  - the same push is more often seen as "violent" when the person pushing is taken to be Black;
  - people in high-status clothing are more often categorised as white.

Prosecutors: thin and thick race perception



Back to the example

A Black arrestee with 2 prior arrests. Which white arrestee is the right comparison?

- **Thin view:** a white arrestee with 2 prior arrests. (Only the phenotypic features change.)
- **Thick view:** a white arrestee with (say) 1 prior arrest. (Number of arrests shapes race perception.)
- Why? With 2 prior arrests, the white counterfactual arrestee would be perceived to have an *unexpectedly* high number of arrests, while the black version would not (cf. p. 256).
- Hu & Kohler-Hausmann: the choice of the comparison is partly **normative**: given which features do we think equal treatment is owed?

References I

Bickel, P. J., Hammel, E. A., & O'Connell, J. W. (1975). Sex bias in graduate admissions: Data from Berkeley. *Science*, *187*(4175), 398–404.

Hu, L., & Kohler-Hausmann, I. (2025). What is perceived when race is perceived and why it matters for causal inference and discrimination studies. *Law & Society Review*, *59*(2), 239–264.

Morgan, S. L., & Winship, C. (2014). *Counterfactuals and causal inference: Methods and principles for social research* (2nd ed.). Cambridge University Press.

Pearl, J. (2009). *Causality* (2nd ed.). Cambridge University Press.